

can be verified only with the public key. With the development of public-key systems, secrecy can be maintained without having to keep track of a large number of keys or sharing a common master-key. If, say, Alex wants to communicate with Neil, she first generates her public/private key pair and sends the public key to Neil over a non-secure channel. Neil uses that key to encrypt information and sends it back to Alex. Only Alex has the private key with which she can decrypt the information. Anyone who intercepts the public key or the encrypted data can't decrypt the message due to the protocols followed during information transfer.

Secrecy in storage

Storage encryption refers to the application of cryptographic techniques on data, both during transit and while on storage media. Storage encryption is gaining popularity among enterprises that use storage area networks (SANs). Secrecy in storage is maintained by storing data in encrypted form. The user has to provide the key to the computer only at the beginning of a session to access the data and it then takes care of encryption and decryption throughout the course of normal use. Hardware devices can also be used for PCs to automatically encrypt all information stored on disk. When the computer is turned on, the user must supply a key to the encryption hardware. The information is plain gibberish without its key thus preventing misuse if the disk is stolen.

Multiple ciphers can be used for individual files and folders. The ciphers and keys should be changed frequently to ensure security of data. However, if the user forgets a key, all of the information encrypted with it makes no sense and is rendered useless. This is why backups of encrypted information are advised to be stored in plaintext. The data is only encrypted while in storage, not when in use. This leaves a loophole for the attackers. The system is vulnerable to a security breach if the encryption and decryption are done in software, or if the key is stored somewhere in the system.

Integrity in transmission

We can use cryptography to provide a means to ensure that data is not altered during transmission, i.e. its integrity is preserved. In electronic funds transfer, it is very important that integrity be maintained. A bank can lose millions if a transaction is illicitly intercepted. Cryptographic techniques are employed to prevent accidental or intentional modification of data during transmission, leading to erroneous actions. One of the ways